600(b)(95) of Regulation NMS,[99] must have access to the protected bid and protected offer. While these rules are impacted by the new definition of round lot, and the data that must be processed and displayed will change as the definition of round lot changes, the minimum data requirements associated with these rules are not changing.[100] Additionally, market participants will need to obtain regulatory data to meet regulatory obligations and to be informed of trading halts, price bands, or other market conditions that may affect their trading activity.[101]

Best execution analysis varies depending upon the characteristics of customers and orders handled. For example, the data requirements for an institutional broker's smart order router ("SOR") executing large algorithmic orders are likely different than for a small retail broker's visual display for non-professional individual investors. Given the large array of potential scenarios, the Commission cannot specify the data elements that may be relevant to every specific situation. Rather, broker-dealers must perform a best execution analysis to determine what data is relevant to obtaining best execution of customer orders, in a manner that is similar to decisions they must make today regarding whether to obtain data content that is available on a proprietary basis.

In addition, the decentralized consolidation model will change the method by which market data is disseminated by introducing competing consolidators, who will offer consolidated market data products, which broker-dealers may choose as a source of market data. The speed of execution, the availability of accurate information affecting choices as to the most favorable

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99 17 CFR 242.600(b)(95).

100 See Proposing Release, 85 FR at 16743–46; infra Section II.D.2(b).

101 See Proposing Release, 85 FR at 16760; infra Section II.H.

market center for execution, and the availability of technological aids to process such information may be relevant factors in conducting a best execution analysis.[102] While all competing consolidators will offer consolidated market data products, they may do so at different prices or at different latencies or with different amounts of data content.[103] Therefore, the selection of a competing consolidator may also be relevant to a broker-dealer's ability to achieve and analyze best execution. Competing consolidators will be required to disclose information about their consolidated market data products, including the services they will offer, the prices for such services as well as performance metrics.[104] These disclosures should help to facilitate a broker-dealer's ability to achieve and analyze best execution because they provide information regarding the timeliness, completeness, and accuracy of the market data offered by competing consolidators.[105] These disclosures also provide statistics on capacity, network delay, and latency, offering additional insight into the technical capabilities and expected performance of a competing consolidator. This information will assist a broker-dealer in selecting an appropriate competing consolidator, which will affect the broker-dealer's ability to obtain "the most favorable terms reasonably available under the circumstances" for its customer orders. The Commission believes that a broker-dealer that uses low-latency or content-rich consolidated market data, whether self-aggregated or received from a competing consolidator, for its proprietary trading, would also be expected to use those data products when pursuing the best

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102 See supra notes 89 and 90 and accompanying text.

103 See infra notes 897, 907–908 and accompanying text.

104 See infra Section III.C.8(c).

105 See supra note 90 and accompanying text.

execution of customer orders, particularly those handled within the same aggregation unit that conducts proprietary trading. For example, a broker-dealer should not use a separate, less performant data source for its customer orders than the data source used for proprietary orders that may interact with those customer orders in a manner disadvantageous to those customer orders.[106]

II. Enhancements to NMS Information

A. Introduction

Today, most market participants utilize electronic trading systems to execute orders for themselves and for their customers. These electronic trading systems, which consume many pieces of data in an effort to trade competitively and efficiently in today's markets, are designed to analyze more information than is provided by the exclusive SIPs. Given that the current market is vastly different from when the national market system was established in the 1970s, the Commission believes that a broad cross-section of market participants would benefit from information that goes beyond SIP data to trade competitively and efficiently and that the information that is provided within the national market system needs to be augmented with new information elements. As discussed in detail below, the Commission is adopting new rules and amending certain existing rules under Regulation NMS to add new elements to the information that is collected, consolidated, and disseminated under the national market system.

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106 Cf. FINRA, Regulatory Notice 15-46, supra note 94. See also letter from Tyler Gellasch, Executive Director, Healthy Markets Association, to Vanessa Countryman, Secretary, Commission, dated May 26, 2020, ("Healthy Markets Letter I") at 4–5; letter from Marcia E. Asquith, Executive Vice President, Board and External Relations, Financial Industry Regulatory Authority, Inc., to Vanessa Countryman, Secretary, Commission, dated May 26, 2020, ("FINRA Letter") at 6.

By way of example, in the 1970s, trading volume in any given stock was concentrated on its listing exchange and trading largely occurred manually with individuals representing orders on exchange floors.[107] Since then, technology has fundamentally altered market operations and trading today largely occurs electronically with little human intervention.[108] Numerous other changes have also impacted how trading occurs. For example, in 2001, decimalization reduced the increment of trading from fractions to pennies and resulted in a reduction in the size of liquidity at the best prices, commonly referred to as the "top of book."[109] The reduction in displayed order interest at the best bid or offer means liquidity is layered across multiple price levels, which makes depth of book information necessary for many market participants and trading systems to trade in an informed and effective manner.

In addition, individual odd-lot quotations, especially in high share price stocks, have become more prevalent[110] and important to market participants as individual share prices have increased.[111] Finally, an increasing proportion of total trading volume is executed during opening and closing auctions, which has made information about orders participating in auctions increasingly important to many market participants. These changes have led market participants to call for additional information to be included in consolidated market data so that market

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107 See Proposing Release, 85 FR at 16728.

108 See id.

109 See id. at 16751.

110 See infra note 240. See also Proposing Release, 85 FR at 16739.

111 See Proposing Release, 85 FR at 16739 (stating that between 2004 and 2019, the average price of a stock in the Dow Jones Industrial Average nearly quadrupled).

participants can participate more fully and competitively.[112] However, very few adjustments[113] have been made to NMS information to account for these changes since the adoption of the 1975 Amendments.

The Commission believes that the content of current SIP data and the mechanism by which SIP data is collected, consolidated, and disseminated has not kept pace with market developments. Therefore, the Commission is adopting these amendments to specify additional information that must be made available pursuant to the effective national market system plan(s).[114] Information about better priced orders in smaller sizes can improve investors' ability to trade at the best prices available. Further, certain market participants can more efficiently place larger sized orders that may not be fully executed at top of book prices using information about the prices of orders outside of the best bids and best offers, and they can more effectively participate in exchange auctions using relevant information about the trading interest in such auctions. Finally, market participants also need to have, and will continue to receive, regulatory

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112 See id. at 16740 (noting multiple Roundtable panelists and commenters supported the addition of odd-lot information to SIP data), 16751–52 (noting multiple Roundtable panelists and commenters supported the addition of depth of book data to SIP data), 16758 (noting multiple Roundtable panelists and commenters supported the addition of auction information to SIP data).

113 See, e.g., Securities Exchange Act Release Nos. 70793 (Oct. 31, 2013), 78 FR 66788 (Nov. 6, 2013) (order approving Amendment No. 30 to the UTP Plan to require odd-lot transactions to be reported to consolidated tape); 70794 (Oct. 31, 2013), 78 FR 66789 (Nov. 6, 2013) (order approving Eighteenth Substantive Amendment to the Second Restatement of the CTA Plan to require odd-lot transactions to be reported to consolidated tape).

114 Section 11A(c)(1)(B) of the Exchange Act provides the Commission with the authority to, among other things, assure the fairness and usefulness of the form and content of quotation and transaction information.

information, administrative data, and other important information to participate effectively in the markets. The Commission received comments on each of these issues.

As discussed more fully below, some commenters, stating that the information is not necessary for all investors, questioned the need to add new information elements.[115] While the Commission recognizes that different market participants need differing amounts of information to meet different trading objectives, the Commission believes that the availability of the new information will enhance the ability of market participants to trade competitively and efficiently and will indirectly benefit investors who place orders in the national market system even if they do not directly consume all of the new data elements by facilitating executing broker-dealers' access to information.[116] In today's market, information about odd-lot quotations, depth of book quotations, and auction information has become highly relevant. Together, these pieces of information can be significantly beneficial in facilitating informed trading decisions, and the Commission believes that they should be more widely distributed and more readily accessible. The Commission anticipates that a variety of consolidated market data products will be developed to meet the various needs investors have for data.[117] The Commission believes that

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115 See, e.g., letter from John A. Zecca, Executive Vice President, Chief Legal Officer, and Chief Regulatory Officer, Nasdaq, to Vanessa Countryman, Secretary, Commission, dated May 26, 2020, ("Nasdaq Letter IV") at 31–34; letter from Elizabeth K. King, Chief Regulatory Officer, ICE, and General Counsel and Corporate Secretary, NYSE, to Vanessa Countryman, Secretary, Commission, dated June 1, 2020, ("NYSE Letter II") at 3–8; letter from Joseph Kinahan Managing Director, Client Advocacy and Market Structure, TD Ameritrade, to Vanessa A. Countryman, Secretary, Commission, dated June 1, 2020, ("TD Ameritrade Letter") at 4.

116 See infra Section II.C.2(a).

117 See infra Section III.E.2(e).

the amendments will enhance the usefulness of NMS information and thus better inform trading and investment decisions for all investors, which in turn will help maintain fair and efficient markets as well as facilitate best execution of customer orders.[118]

Accordingly, as discussed in more detail below, the Commission is adopting several new defined terms under Rule 600 of Regulation NMS to specify, and as a result expand and enhance, the data that Regulation NMS requires to be collected, consolidated, and disseminated. Importantly, the Commission is adopting two new definitions under Regulation NMS—"consolidated market data" and "core data"—to specify the components of NMS information that are required to be collected, consolidated, and disseminated under the national market system. "Consolidated market data product" is defined as any data product developed by a competing consolidator that contains consolidated market data or any of the elements or subcomponents thereof. The Commission is also adopting additional defined terms to further set forth the scope of information included within the definitions of consolidated market data and core data. The definitions include information that is currently provided by the exclusive SIPs as well as new information designed to ensure that brokers, dealers, and investors have available information with respect to quotations for and transactions in securities that is prompt, accurate, reliable, and fair.[119]

B. Definition of "Consolidated Market Data" under Rule 600(b)(19)

1. Proposal

The Commission proposed to expand the content of the NMS information that would be required to be collected, consolidated, and disseminated under the rules of the national market

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118 See supra Section I.E (discussing the implications for best execution).

119 See supra Section I.A.

system through the proposed definition of "consolidated market data." Specifically, the Commission proposed that consolidated market data would include the following data, consolidated across all national securities exchanges and national securities associations: (1) core data; (2) regulatory data; (3) administrative data; (4) exchange-specific program data; and (5) additional regulatory, administrative, or exchange-specific program data elements defined as such pursuant to the effective national market system plan or plans required under Rule 603(b).[120] In addition, the proposed definition of consolidated market data would be used to delineate the responsibilities and obligations of the SROs under Rule 603(b) and competing consolidators under Rule 614. These rules implement the decentralized consolidation model, which is discussed in more detail in Section III below.

2. Final Rule and Response to Comments

The Commission received a number of comments on the proposed expansion of NMS information related to the specific elements that make up consolidated market data,[121] and the Commission also received some comments on the proposed definition of consolidated market data. One commenter supported the expansion of NMS information to include the proposed elements of consolidated market data.[122] Another commenter agreed with the proposed definition, stating that these data elements need to be clearly defined and categorized and that

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120 As discussed below, the Commission also proposed and is adopting definitions for "core data," "regulatory data," "administrative data," and "self-regulatory organization-specific program data." See infra Sections II.C, II.H, II.J, II.K, respectively.

121 See infra Sections II.C through II.K.

122 See Capital Group Letter at 2.

"tight definitions would assist to 'preserve the integrity and affordability of the consolidated data stream.'"[123]

Other commenters, however, stated that the Commission should allow additional core data elements to be included in consolidated market data through a process other than Commission rulemaking.[124] A different commenter stated that the proposed changes to consolidated market data are "not appropriately tailored to the needs of the market" and "are overly broad and unnecessarily complex."[125] Another commenter, while agreeing that the definition of consolidated market data should be defined as proposed, suggested that it should only include depth-of-book data, certain odd-lot information, and three options for including auction data.[126]

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123 TD Ameritrade Letter at 3 (quoting Regulation NMS Adopting Release).

124 See Clearpool Letter at 11 (stating that the Commission should provide flexibility in the definition of core data or the process by which the elements of core data are determined); RBC Letter at 4 (stating that the proposed definition of core data should serve as a "floor" that the Operating Committee should be permitted to expand upon (but not reduce) pursuant to Plan amendments); letter from Emil R. Framnes, Global Head of Trading, and Simon Emrich, Market Structure and Trading Research, Norges Bank Investment Management Letter, to Vanessa Countryman, Secretary, Commission ("NBIM Letter") at 5 ("[I]t might be prudent to allow for further modification of the definition of core data as market structure evolves.").

125 NYSE Letter II at 3.

126 See letter from Kelvin To, Founder and President, Data Boiler Technologies, LLC, to Vanessa Countryman, Secretary, Commission, dated May 26, 2020, ("Data Boiler Letter I") at 19–20. However, this commenter also stated that administrative data should be included in the proposed definition of consolidated market data. See id. at 34.

The Commission is adopting the definition of consolidated market data largely as proposed.[127] As discussed in detail below,[128] the Commission continues to believe that expanding the NMS information that is required to be provided under the rules of the national market system, as set forth in the definition of consolidated market data, would support more informed trading and investment decisions by market participants in today's markets and facilitate the best execution of customer orders by the full range of broker-dealers.[129] As reflected in comments received from a variety of market participants, each of the elements of consolidated market data—and in particular the expansion of core data to include quotation interest in smaller orders of higher-priced stocks, depth of book data, and auction information—would provide significant, useful information to market participants.[130] Consistent with the views of market participants—many of whom will be the users of consolidated market data—that this data would be useful to them to improve investment decisions and facilitate the best execution of customer orders, the Commission believes that the definition of consolidated market data is "appropriately tailored" to market participants' needs, that it is not overly broad, and that it does not entail unnecessary complexity.[131] In addition, the proposed decentralized consolidation model permits competing consolidators to offer, and market participants to

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127 The Commission is modifying the definition of exchange-specific program data to be self-regulatory organization-specific program data. See infra Section II.K.

128 See infra Sections II.C through II.K.

129 See Proposing Release, 85 FR at 16735. See also infra Section II.C.2(a); supra Section I.E.

130 See infra Sections II.C through II.K.

131 See NYSE Letter II at 3.